The Madras government, in a public citation dated October 1838, recognized Captain Cotton's development of a 'rotatory' engine. Captain Cotton left on sick leave for Tasmania on 12 December 1838. He returned to India in October 1840 briefly, and then finally in April 1843 for good. During an experiment with his engine soon after his arrival in Hobart in 1838, Cotton had been seriously injured by an explosion of the boiler. He was unable to walk for months. His influence in Tasmania helped in irrigation gaining acceptance as an aid to cereal production and the livestock industry, and for developing various apparatus for the same, including a steam-driven digging machine. [20]

The Red Hills Railway was left incomplete and was eventually shut down after Captain Cotton left, but the experiments stated above gave him an important insight into building and running railway lines, helping him in the completion of the canal work along the Godavari .

'Circumstances arose, which prevented the completion of the line, but it afforded practical experience, which was of considerable use to him in the execution of the Godavari works years after, where railway lines played an important part in facilitating the prompt construction of the great dams,' quotes his daughter in his biography. [21]

Captain Cotton eventually returned to India and the Madras Corps of Engineers, and by mid-1844, had been given charge of irrigation works on the Godavari, where a large railway was built under his supervision to facilitate the work.

Porto Novo Iron Works became the Indian Steel and Iron Company in 1833 and suffered severe losses. It became the East India Iron Company in 1853—formed primarily by merchants, and the efforts to revive the company continued until they were finally given up in 1874, when the company went into liquidation. [22]

ROORKEE'S LOCOMOTIVE THOMASON

Another experiment that failed but remains documented in India's railway history is the arrival of the locomotive Thomason. This took place in another part of the country, about two decades after the Bombay experiments, but before the first official train started running. Its story began in north India, in 1851. Plans for regular railway lines in India had